

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8392**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CSA13
Exam Type:	Written	Block:	305	Seat No:	2022700059
		Course Section:	1		

L 41,672,986, 1,1505, 41485



Name of the candidate. Prachi Anil Tayshete

Candidate's UID number:

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Examination class room number:

3	0	5
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Invigilator's signature with date:

 - 8/10
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(To be filled in by the candidate)

EXAMINATION: BE BTech
(For example FY MCA/FY BTech)

PROGRAM: CSE-D5

DAY AND DATE: 08/10/2025

COURSE NAME: Deep Learning

SEMESTER :

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

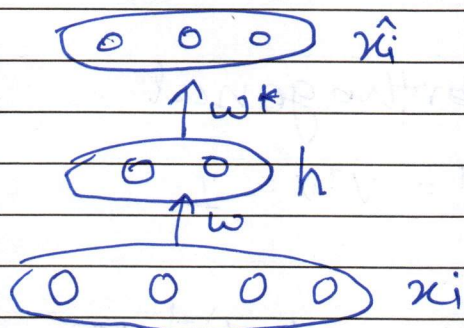


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Q.2b



- Autoencoder takes a sequence x_i as input, decodes to h & decodes to $\hat{x}_i$.

- Undercomplete: $\dim(h) < \dim(x_i)$
Thus, the autoencoder has to learn specific rules for updating the parameters & minimising the loss.

$$\text{loss} = \frac{1}{2} \| \hat{x}_i - x_i \|^2$$

$$\hat{x}_i = HW^* \quad \text{let } x_i = X.$$

For minimising loss,

$$X = HW^* \quad \text{--- (1)}$$

$$X = U\Sigma V^T \quad \text{--- (2) (SVD provides soln)}$$

$$HW^* = U\Sigma V^T$$

One possible solution is,

$$H = U\Sigma$$

$$W = V^T$$

Let's prove this

$$H = U\Sigma$$

$$H = (XX^T)(XX^T)^{-1} U\Sigma$$

$$\dots (A \cdot A^T = I = 1)$$

$$\text{Now, } X = U\Sigma V^T$$

$$H = (X(U\Sigma V^T)^T)(U\Sigma V^T(U\Sigma V^T)^T)^{-1} U\Sigma$$

$$(AB)^T = B^T A^T$$

$$H = (XV^T \Sigma^T U^T)$$

$$H = (XV^T \Sigma^T U^T)(U\Sigma V^T V^T \Sigma^T U^T)^{-1} U\Sigma$$

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U and V are orthogonal matrices.

$$U^{-1} = U^T \quad VV^T = V^TV = I$$

$$V^{-1} = V^T$$

$$\therefore H = XV \Sigma^T U^T (U \Sigma \Sigma^T U^T)^{-1} U \Sigma$$

$$(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$$

$$\therefore H = XV \Sigma^T U^T \underline{(U^T)^{-1}} \underline{(\Sigma \Sigma^T)^{-1}} \underline{U^{-1} U \Sigma}$$

$$H = XV \underline{\Sigma^T} \underline{(\Sigma^T)^{-1}} \underline{\Sigma^{-1} \Sigma}$$

$$H = XV$$

$\therefore H$ is a linear encoding matrix of X .

$$W^* = V^T \text{ (eigen values of } V^T)$$

In PCA, we normalise,

$$x_{ij} = \frac{1}{\sqrt{m}} \left(\hat{x}_{ij} - \frac{1}{m} \sum x_{ij} \right)$$

(subtract mean)

$$X = \frac{1}{\sqrt{m}} X'$$

$$XX^T = \frac{1}{m} (X') (X')^T \text{ (Covariance matrix whose eigen values are found)}$$

Thus autoencoder links with PCA $\rightarrow$

① linear encoder

② linear decoder

③ loss function = MSE.

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- For an overcomplete autoencoder, $\dim(h) \geq \dim(x_i)$
- Encoded output is bigger than input.
- Instead of learning some parameters, it just copies input to output. (identity).
- It loses its ability to generalise and overfits. It performs very poor on unseen inputs.
- L2 Regularisation adds a penalty term to the loss function to improve generalisation.

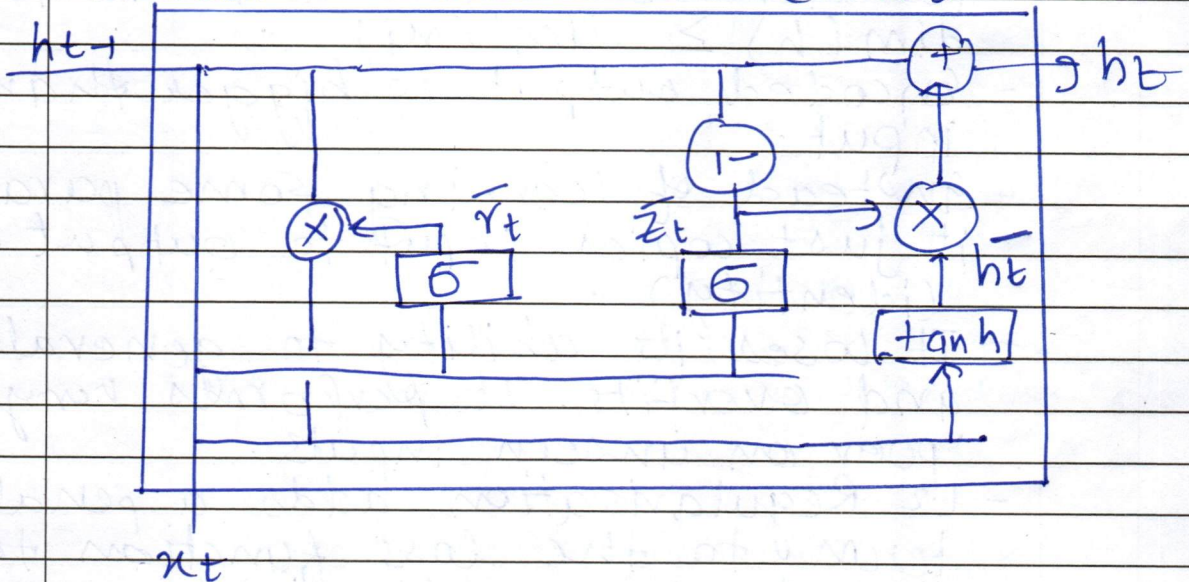
$$L = \frac{1}{n} \sum_i \sum_j (\hat{x}_{ij} - x_{ij})^2 + \lambda \|\theta\|^2$$

- This corrupts the input & improves learning.

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Q.3b - Gated Recurrent Unit (GRU) -



① Reset Gate: it tells how much of previous information is to be remembered / forgotten.

$$\bar{r}_t = \sigma[W_r(x_t \cdot h_{t-1})]$$

② Update Gate: How much of new info is to be added.

$$\bar{z}_t = \sigma[W_z(x_t \cdot h_{t-1})]$$

③ Candidate state: It is determined by the current input and previous hidden state.

$$\bar{h}_t = \tanh[W_h(\bar{r}_t \cdot x_t \cdot h_{t-1})]$$

④ Final hidden state: It is weighted average of previous hidden state and candidate hidden state.

$$h_t = (1 - \bar{z}_t)h_{t-1} + \bar{z}_t\bar{h}_t$$

↑
Convex combination of previous hidden state and candidate hidden state.



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- LSTMs use 3 gates $\Rightarrow$

① Forget gate :

$$f_t = \sigma[w_f(x_t, h_{t-1}) + b_f]$$

- f_t is multiplied with C_{t-1} (cell state)
- It tells how much info is to be forgotten.

② Input Gate :

[Potential mem. to remember]

$$i_t = \sigma[w_i(x_t, h_{t-1}) + b_i]$$

$$\bar{C}_t = \tanh[w_c(x_t, h_{t-1}) + b_c]$$

$\hookrightarrow$ [Potential long term memory]

$$C_t = i_t \times \bar{C}_t + C_{t-1}$$

③ Output gate : Determines new state

$$O_t = \sigma[w_o(x_t, h_{t-1}) + b_o]$$

$$h_t = O_t \cdot \tanh[C_t] \Rightarrow \text{new hidden state.}$$

- GRUs use only one update gate in place of input and output gates of LSTM.
- GRUs use 2 gates while LSTM use 3 gates.
- GRUs compute faster as compared to LSTMs.
- LSTMs are computationally expensive.
- GRUs are well suited where speed is required.
- But, LSTMs better where we want to preserve more long term memory.

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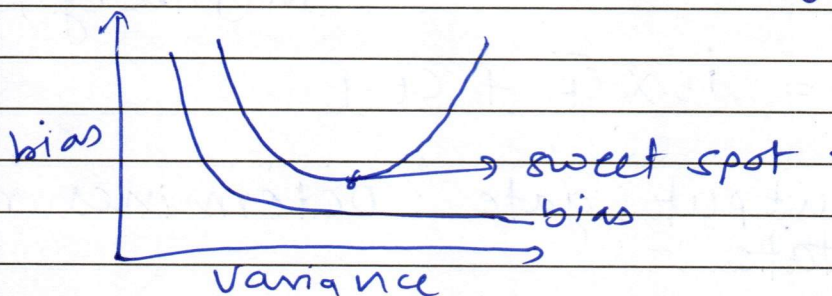
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Q.1b $E(\hat{f}(n)) = \text{Bias}^2 + \text{Variance} + \text{Irreducible error}$

$$E(\hat{f}(n)) = E((\hat{f}(n_i)^2 - f(n_i))) + E((\hat{f}(n_i) - f(n_i))) + \delta$$

Error = Bias² + Variance + Irreducible error

- If model capacity is too high, model is well trained on train data.
- It has low bias and high variance.



- Its variance decreases till its optimal & again increases.
- The model memorises the input data and loses its ability to generalise on unseen data.
- Overfitting, thus, occurs.

As model capacity high.

↳ Bias² → high
↳ Variance → high.

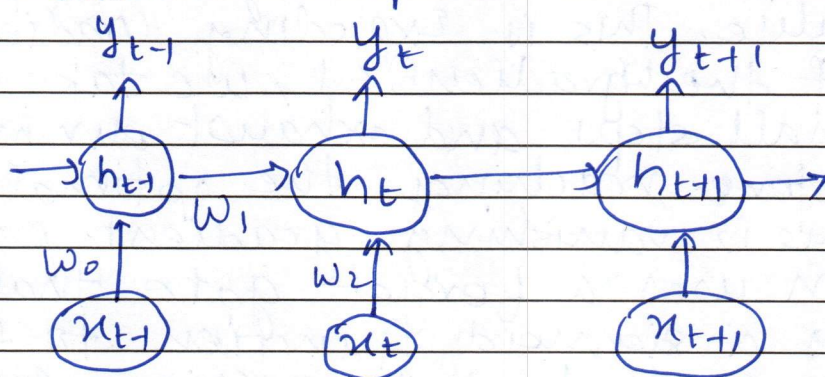
Total error → high.

- Thus, overfitting is caused by high variance & model loses its ability of generalisation which can be solved by regularisation.

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Q.3a - RNN (Recurrent Neural network) -

- It has internal memory.
- It works well for sequential data.
- It considers previous output as well as current input to calculate current output.



$$h_t = f_c [x_t, h_{t-1}]$$

Consider sigmoid function,

$$h_t = \sigma [x_t, h_{t-1}]$$

$$L = \frac{1}{2} \| y_i - \hat{y}_i \|^2$$

$$L = \frac{1}{2} \| y_i - (\sigma(x_t, h_{t-1})) \|^2$$

$$\frac{\partial L}{\partial h_{t-1}} = \frac{1}{2} \cdot 2 \| y_i - (\sigma(x_t, h_{t-1})) \| \cdot \sigma'(x_t, h_{t-1})$$

$$\text{Now, } \sigma'(z) = \sigma(z)(1 - \sigma(z))$$

$$\text{Let } z = x_t, h_{t-1}$$

$$\frac{\partial L}{\partial h_{t-1}} = \| y_i - \sigma(z) \| \cdot \sigma(z) \cdot [1 - \sigma(z)]$$

Required expression.

$\sigma(z)$ is a function of weights & biases. h_{t-1} also depends on weights.

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- Thus, Gradient depends on weights.

- In RNN,

① If the $w(\text{gradient})$ is > 1 , over a large number of steps, the value increases rapidly & we keep jumping / oscillating from one peak to next missing the optimal value. This is exploding gradient.

② If the $w(\text{gradient}) < 1$, we take very small steps and exhaust our model before reaching the optimal value. This is vanishing gradient problem.

- LSTM uses a forget gate that uses a sigmoid function to tell how much of this previous long term memory is to be remembered.

$$f_t = \sigma[w_f[x_t, h_{t-1}] + b_f]$$

- LSTM uses the concept of maintaining a cell state separately. The basic intuition is to store long term dependencies separately from short term memory.

- Further no weights are added to this cell state C_t that shows long term memory.

- This way, vanishing / exploding gradients managed.

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Q2a. - In convolution, we slide a filter $f \times f$ over input $w_1 \times h_1 \times d_1$.

- No. of filters = k

- Output size = $w_2 \times h_2 \times d_2$

s = Strides

$$d_2 = k$$

$$w_2 = \frac{w_1 - f}{s} + 1$$

$$h_2 = \frac{h_1 - f}{s} + 1$$

- With padding,

$$w_2 = \frac{w_1 - f + 2p}{s} + 1$$

$$h_2 = \frac{h_1 - f + 2p}{s} + 1$$

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$$Q19. f(w) = \frac{1}{2} \|Xw - y\|^2$$

↳ loss function.

$$\frac{\partial f}{\partial w} = \frac{1}{2} 2 \|Xw - y\| \frac{\partial}{\partial w} [Xw - y]$$

$y = \text{constant}$.

$$\frac{\partial f}{\partial w} = \|Xw - y\| \cdot X$$

$$\nabla f = \|Xw - y\| \cdot X$$

$$\Delta \theta \quad \theta_{t+1} = \theta_t - \eta \nabla \theta$$

$$w_{t+1} = w_t - \eta (\|Xw - y\| \cdot X)$$

$$b_{t+1} = b_t - \eta (\|Xw - y\|)$$

for convergence, $P \rightarrow w_i x_i \geq 0$

if $w^T x < 0$,

$$w_{t+1} = w_t + P_i$$

$$\|P_i\| = 1$$

$$\|w^*\| = 1$$

$X^T X = \text{covariance matrix}$.

it has eigen values $\lambda_1, \lambda_2, \dots$

$\lambda_{\max}$ is the maximum value.

$$\eta < \frac{2}{\lambda_{\max}}$$

[illegible]

[illegible]

[illegible]

